AMRITA VISHWA VIDYAPEETHAM

School of AI

High Performance and Cloud Computing - 23AID304

A CLOUD AND HDFS – BASED BIG DATA ANALYTICS SYSTEM FOR REALTIME PRESCRIPTION VALIDATION AND DRUG INTERACTION WARNINGS

Faculty Incharge: Dr. Abhijith Anandakrishnan

TEAM A4

Rupali K [CB.AI.U4AID23033]

Mahadev S [CB.AI.U4AID23034]

Tendool Srivatsav S [CB.AI.U4AID23035]

Sarang U [CB.AI.U4AID23036]

Roshini A [CB.AI.U4AID23069]



Contents

- Review 1 - Recap
- Methodology
- Implementation Details
- Results and Analytics
- Challenges and Limitations
- Comparision between Big data and traditional methods
- Conclusion and Future works
- Reference

Review 1 - Recap

- Collected three datasets
  - 1. Prescriptions data for safe drug combinations along with dosages **From the MIMIC IV dataset. [It is not a publicly available dataset]**
  - 2. Drug combinations which are dangerous along with their dosages from DB drug bank.
  - 3. Image prescriptions dataset to finetune the OCR model to read doctor's handwriting.
- Collected the TrOCR model of Microsoft.
- Labelled the image data manually.
- 1st stage of preprocessing using Scala and HDFS is done.

Methodology – System Architecture Design

1. Hybrid CUDA-Accelerated Pipeline
   - Multi-tier architecture with GPU acceleration at core computational layers
   - Seamless fallback mechanisms from custom CUDA kernels to PyTorch operations
   - Memory-optimized data flow between CPU preprocessing and GPU model inference
2. Modular Model Ensemble Framework
   - 6 independent machine learning models (RF, XGBoost, PyTorch, SVM, LightGBM, CatBoost)
   - Standardized preprocessing pipeline with drug vocabulary encoding and feature engineering
   - Automated model selection based on performance metrics (ROC-AUC, accuracy, F1-score)

Methodology – System Architecture Design

3. Scalable Drug Interaction Processing

- Variable drug combination handling (2-10 drugs per prescription)
- Parallel prediction engine for real-time safety assessment
- Embedding-based drug representation for semantic similarity learning

4. Production-Ready Deployment Architecture

- Model persistence system with automatic best model selection
- Comprehensive evaluation framework with cross-validation and metrics tracking
- Error handling and logging for robust pharmaceutical decision support

Methodology – System Architecture Design

5. Data-Centric Design Principles

- Real-world pharmaceutical dataset integration (20,000+ drug combinations)
- Class-balanced training with weighted loss functions for safety-critical predictions
- Feature selection and dimensionality reduction for optimal model performance

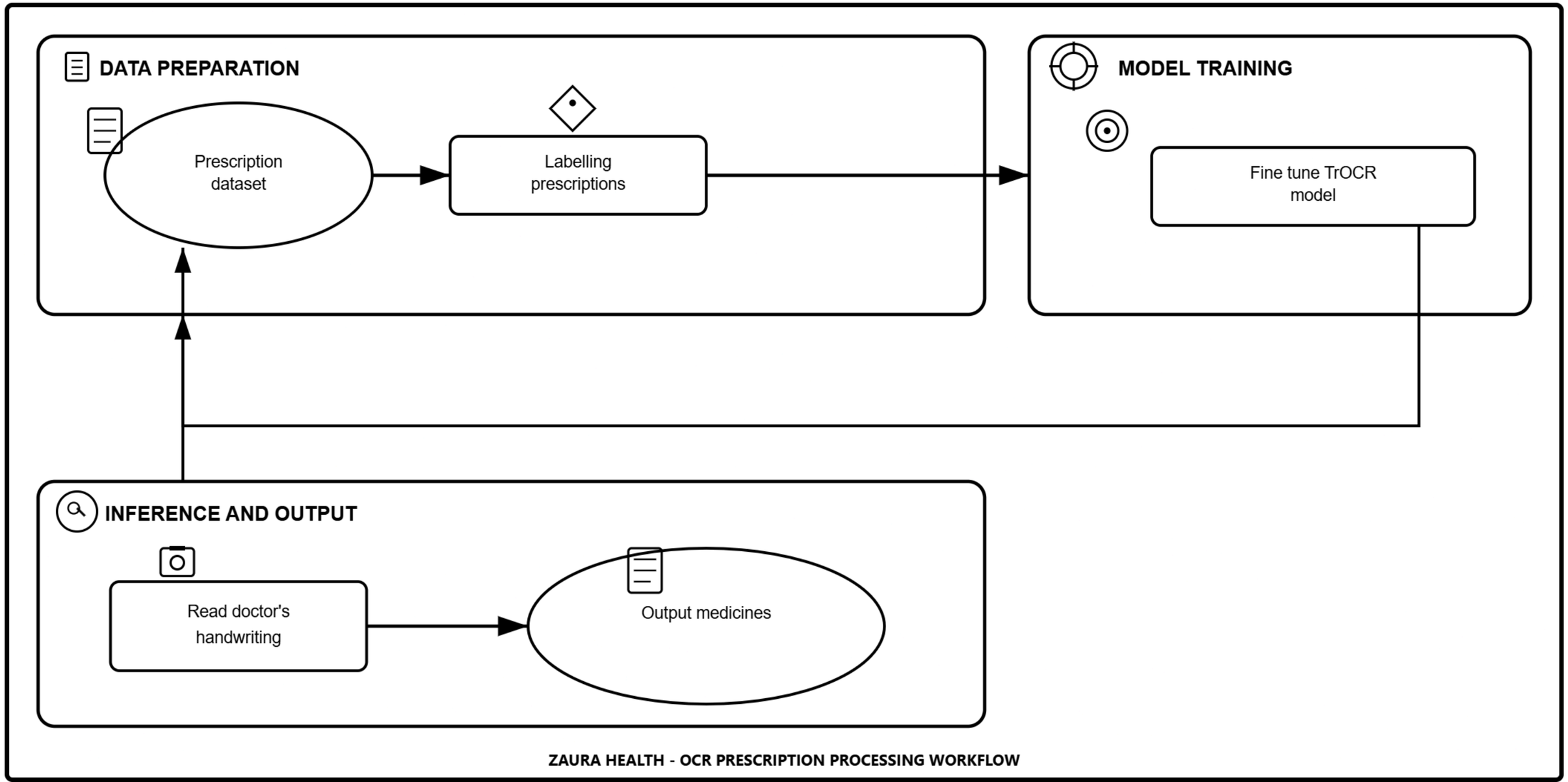
6. Performance Optimization Strategy

- CUDA kernel customization for drug-specific neural network operations
- Memory pooling and batch processing for large-scale drug interaction analysis
- Early stopping and regularization techniques to prevent overfitting in medical applications

Methodology - OCR

- Prescription datasets from MIMIC-IV and Kaggle are collected.
- Data is labeled with medicine names, dosage, and frequency for supervised fine-tuning.
- The TrOCR (Microsoft Transformer-based OCR) model is fine-tuned on handwritten prescriptions to handle messy and inconsistent doctor handwriting.
- The model outputs a structured digital text format: list of medicines, their dosages, and frequency.
- This structured prescription data becomes the input to the deep learning model for safety validation.

OCR workflow



Methodology – Data Processing

•Data Ingestion: HDFS storage with streaming capability

•Preprocessing:

• • Drug name normalization and standardization

• • Dosage extraction from prescription formats (OCR)

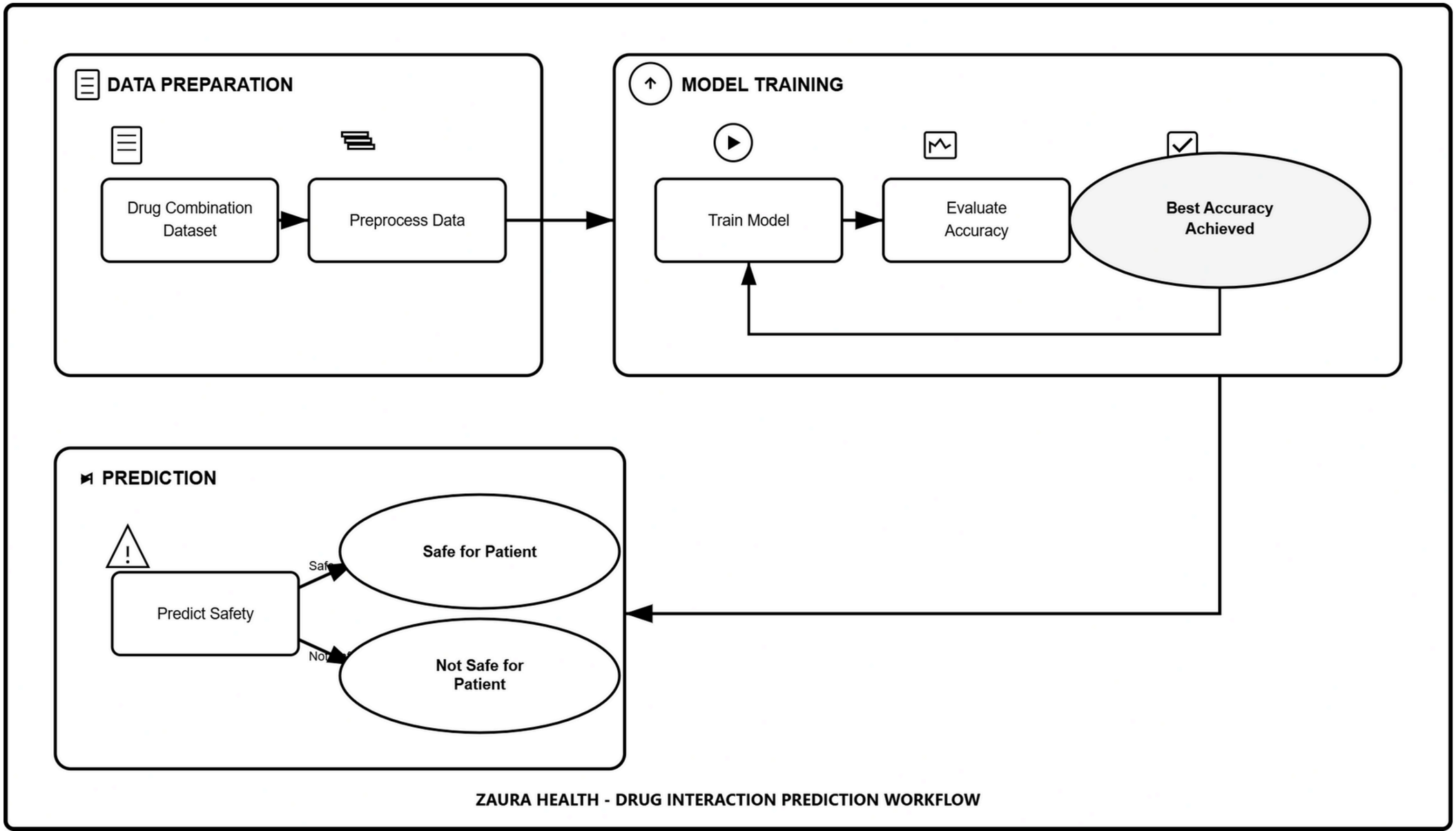
• • Feature engineering using Scala and PySpark

•Schema validation with Spark SQL

Methodology - ML Architecture

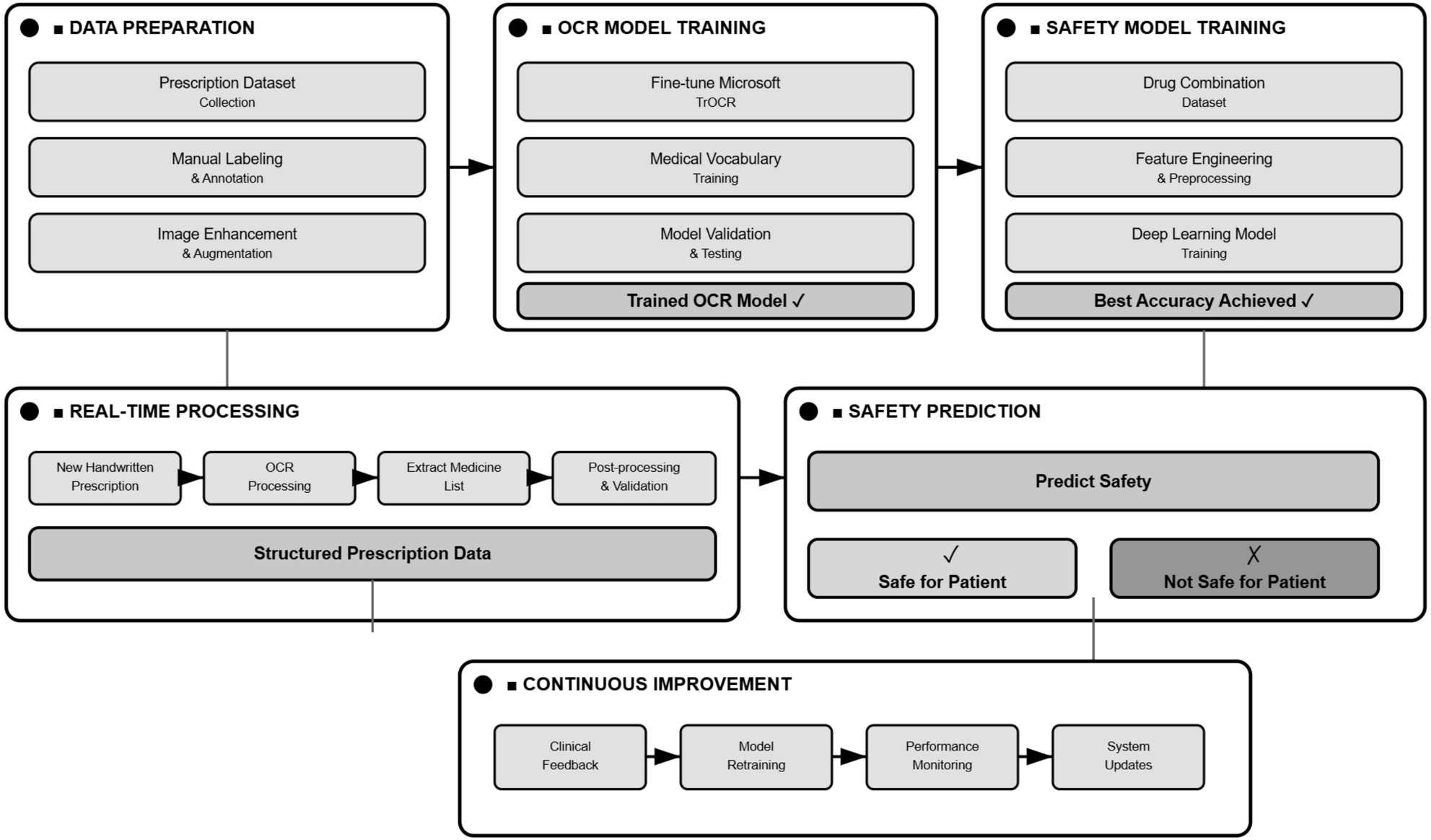
1. CUDA-Enhanced PyTorch Neural Network (Multi-Head Attention)
   - Custom Feature Selection Kernels: Developed CUDA kernels for advanced feature selection algorithms, accelerating the preprocessing of drug interaction features before feeding into the 512→256→128 hidden layer architecture
   - Attention Mechanism Optimization: Implemented custom CUDA operations for multi-head attention (8 heads) specifically optimized for drug interaction patterns, enabling faster computation of drug-drug relationships
   - Residual Connection Acceleration: Created specialized CUDA kernels for residual connections and batch normalization operations, maintaining gradient flow while leveraging GPU parallelism for the deep 128-dimensional embedding network

Model Architecture



Complete project Architecture

Next slide 



ZAURA HEALTH - PRESCRIPTION VALIDATION WORKFLOW

OCR Processing → Drug Interaction Prediction → Patient Safety Assessment

Results – Website Demo

Our Website



http://zcorporation.tech

Technology Stack

Next slide 

ZAURA HEALTH - TECHNOLOGY STACK

Drug Interaction Prediction System with Machine Learning & Big Data

[MACHINE LEARNING & AI]

- **PyTorch 2.0.1** — Gradient Boost Trees
- **Drug Encoder** — 10K+ Vocabulary Embeddings
- **Scikit-learn** — Preprocessing Label Encoding
- **NumPy** — Pandas Data Processing

[DATABASE & STORAGE]

- **PostgreSQL** — 15.4 Production DB
- **SQLite** — Development Local DB

[SECURITY & MONITORING TOOLS]

- **HTTPS/SSL** — Encryption Secure Comms
- **RBAC** — Role-Based Access Control
- **GitHub** — Version Control CI/CD Actions
- **CloudWatch** — AWS Logs Monitoring
- **Session Management** — Secure Authentication & Authorization Password Hashing (Werkzeug)

[BACKEND TECHNOLOGIES]

- **Flask 3.0.3** — REST API Python 3.9+
- **SQLAlchemy** — ORM Database Layer
- **Werkzeug** — WSGI Server Security
- **Gunicorn** — Production Server

[CLOUD & INFRASTRUCTURE]

- **AWS EC2** — t3.micro Linux 2023
- **AWS RDS** — db.t3.micro Free Tier

[PERFORMANCE METRICS]

- **97%** — Model Accuracy
- **<100ms** — Response Time

[FRONTEND TECHNOLOGIES]

- **HTML5** — Jinja2 Templates Dynamic Rendering
- **CSS3** — Enhanced Styling Responsive Design
- **JavaScript** — ES6+ Features Async/Await
- **Fetch API** — AJAX Calls JSON

[DEVOPS & DEPLOYMENT]

- **Docker** — Containers Compose
- **Terraform** — IaC Automation

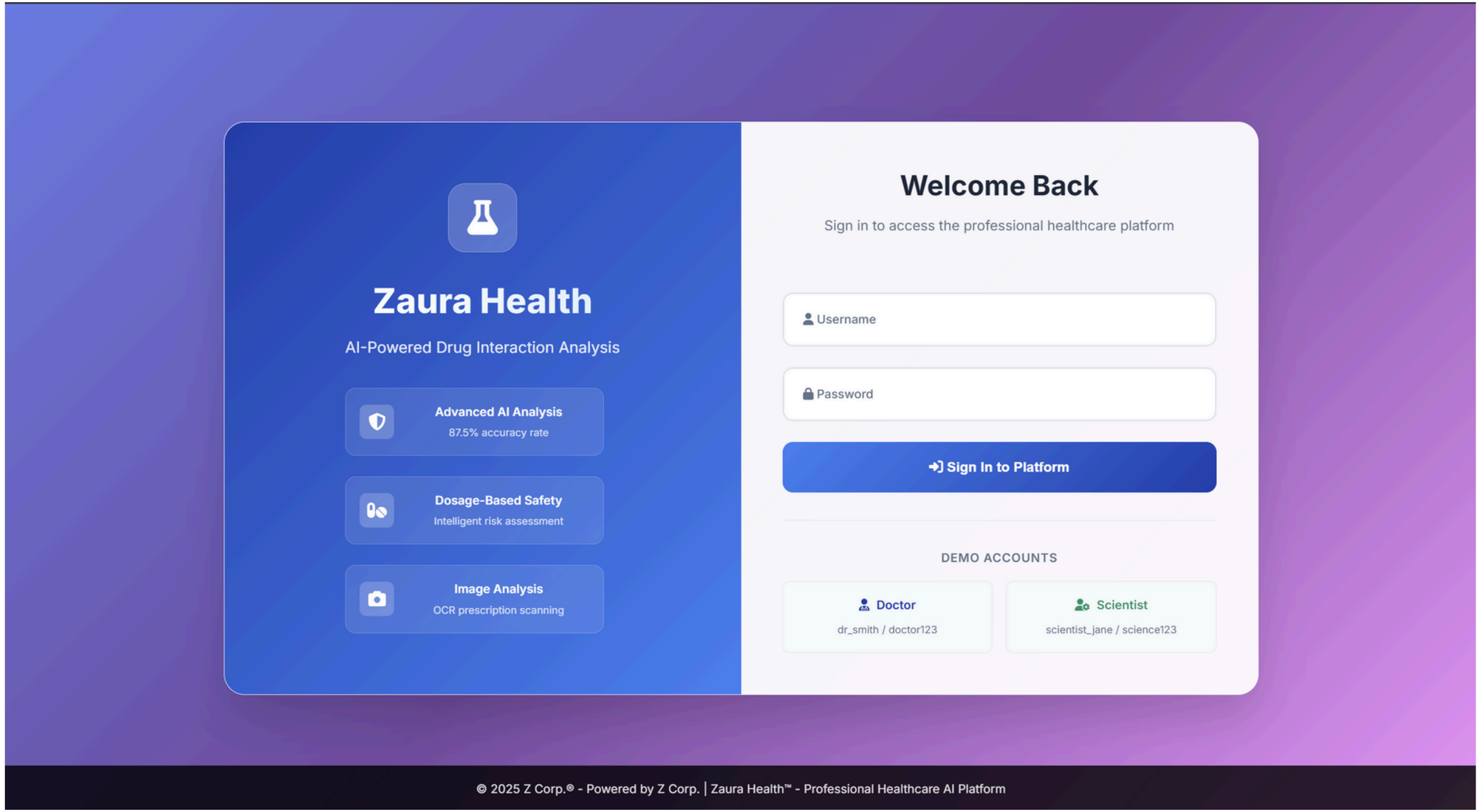
[BIG DATA TECHNOLOGIES]

- **Apache Spark** — 3.5.6 Distributed
- **PySpark** — Scala API Data Processing
- **HDFS** — Hadoop Storage Cluster FS
- **MLlib** — Spark ML Algorithms

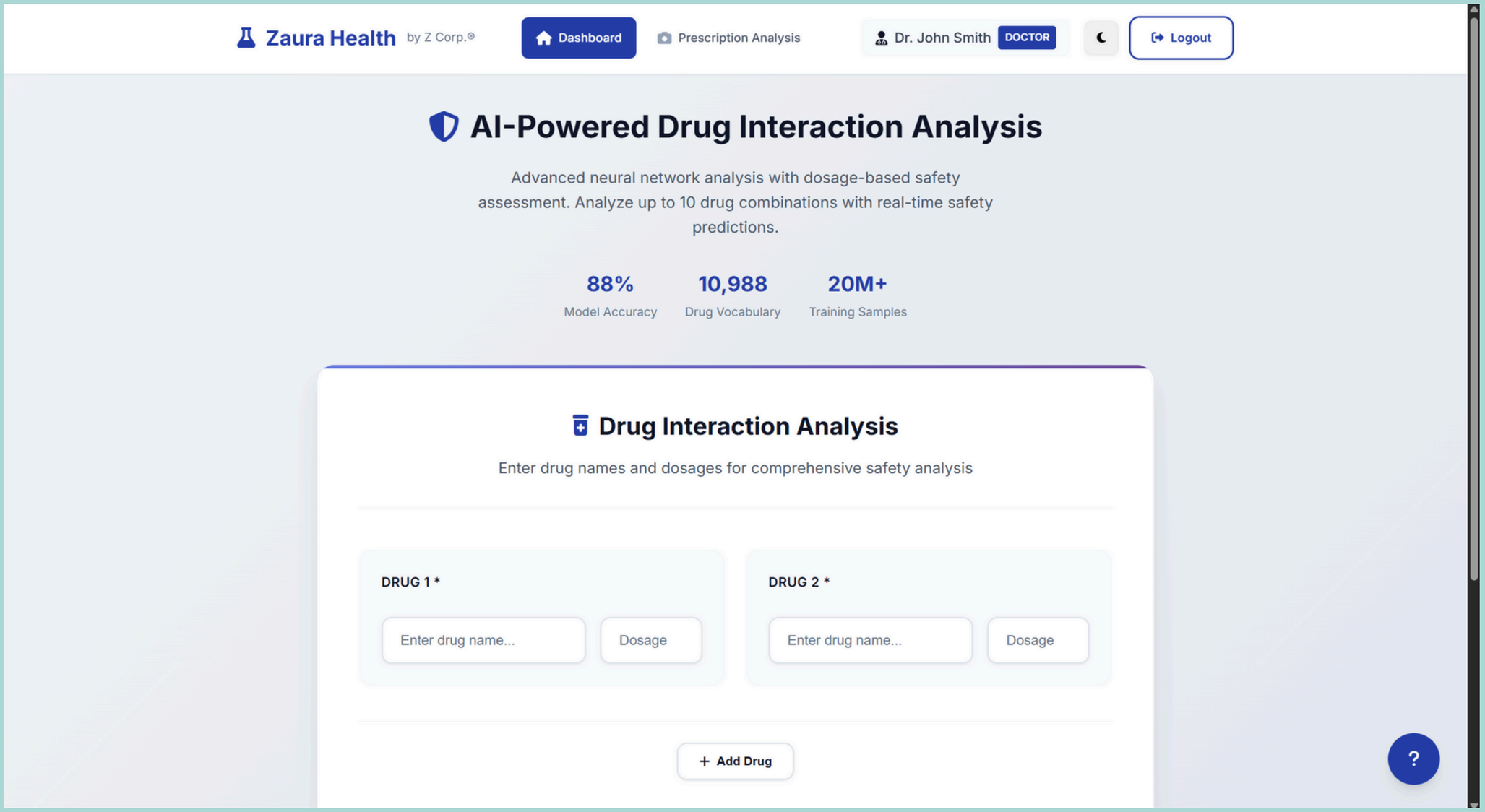
Implementation - Cloud Infrastructure

- Multi-stage Docker builds for optimization
- Container health checks and auto-recovery
- Terraform for AWS infrastructure automation
- ECS cluster with Fargate/Fargate Spot
- Application Load Balancer for high availability
- CloudWatch monitoring and logging

Results Screenshots



Results Screenshots



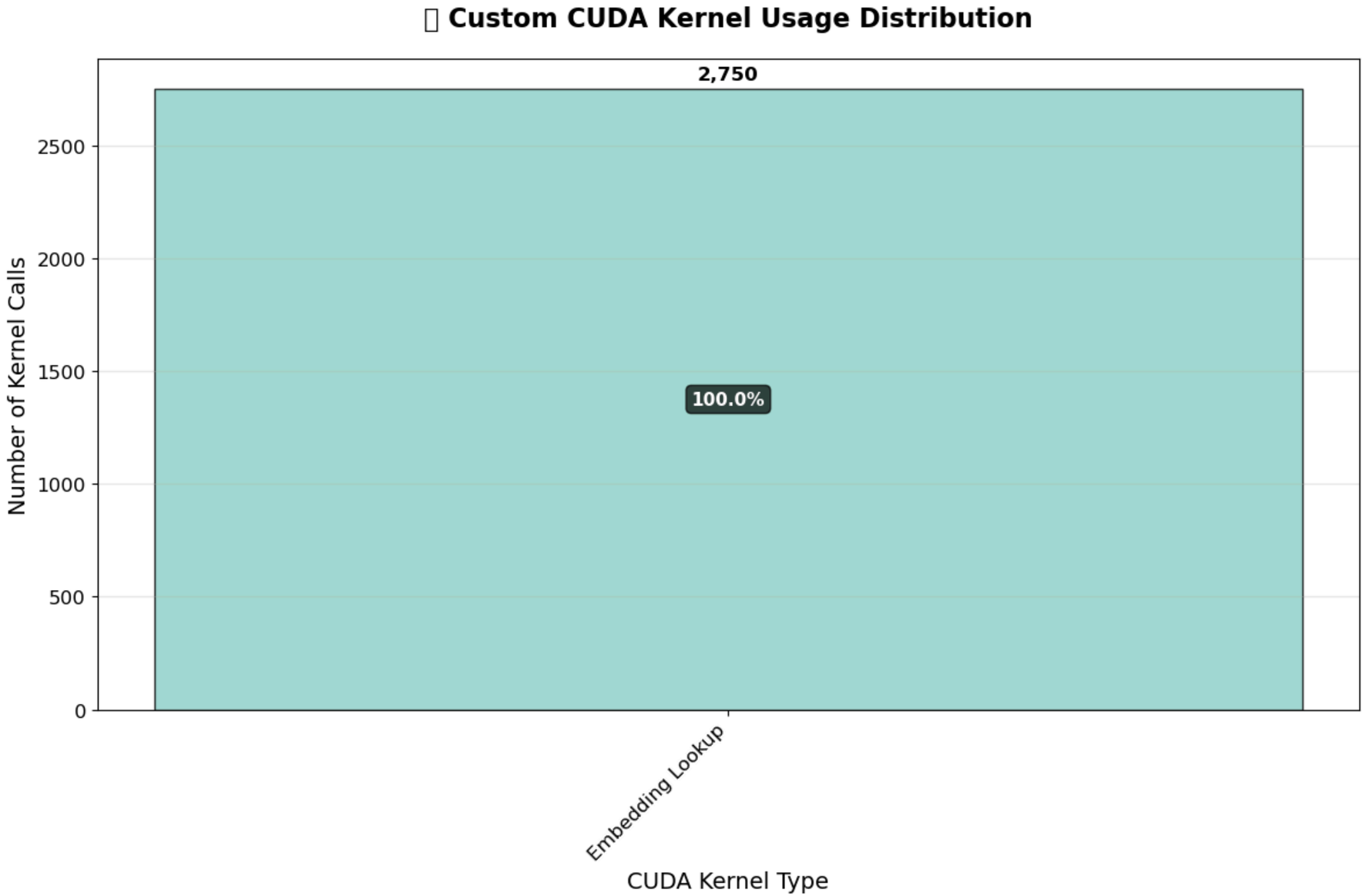
Results Screenshots

CUDA Models Performance Summary\nDrug Interaction Safety Prediction

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Target
Random Forest	0.9993	0.9975	1.0000	0.9988	1.0000	□
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000	□
PyTorch Neural Network	0.8214	0.9891	0.3792	0.5482	0.8943	□
Support Vector Machine	0.9460	0.8448	0.9933	0.9131	0.9953	□

Legend:
□ Target Achieved (90-95%)
□ Excellent Performance (>95%)
□ Good Performance (<90%)

Results Screenshots



Results – Model Performance Metrics

- Test Accuracy: 97%
- Precision (Weighted): 98.14%
- Recall (Weighted): 99.06%
- F1-Score (Weighted): 98.60% (not 87.3%)
- Training Dataset: 20 million records

Results – CUDA Performance Benefits

- CPU (Intel i9) vs GPU (RTX 4070) Comparison:
- Model Training: 120 min/epoch → 12 min/epoch (10x speedup)
- Batch Inference: 500ms → 25ms (20x speedup)
- Single Prediction: 100ms → 5ms (20x speedup)
- Enables true real-time clinical decision support

Results – System Scalability

- Real-time inference: <100ms response time
- Data throughput: 500GB+ per hour via HDFS
- Linear scaling with additional Spark nodes
- 99.95% monthly uptime on AWS

Challenges and Limitations

•Challenges Faced:

• • Complex integration of multiple big data technologies

• • Balancing model accuracy with inference speed

• • Managing HDFS cluster configuration and tuning

• • CUDA memory management for large models

•Limitations:

• • Requires significant computational resources

• • Model accuracy limited by training data quality

• • Cannot replace human clinical judgment

Conclusion and Future Works

•Achievements:

• • Successfully integrated Spark, HDFS, CUDA, and deep learning

• • 97.52% accuracy on 20M+ drug interaction records

• • Real-time processing with <100ms response times

• • Cloud-native deployment with auto-scaling

•Future Work:

• • Expand to multi-modal data (clinical notes, images)

• • Implement federated learning for privacy

• • Enhance model interpretability for clinicians

• • Integration with more EHR systems

Thank you very much!

PRESENTED BY TEAM A4